

Ulugbek Abdullaev

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Education

Purdue University Master of Science in Electrical and Computer Engineering	West Lafayette, IN Aug 2024 - May 2026
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Tashkent State Transport University Bachelor of Science in Automation and Control	Tashkent, Uzbekistan Sep 2019 - Jun 2023
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Experience

Controls Engineering Intern Uzbekistan Railways	Feb 2023 - May 2023 Tashkent, Uzbekistan
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- Conducted testing and troubleshooting of control circuits to ensure operational reliability
- Utilized Microsoft Visio to develop control circuits for train movement safety
- Assisted in the integration of sensors and monitoring devices into control systems to enhance real-time data collection and system responsiveness

Controls Engineering Intern Uzbekistan Railways	Jun 2022 - Aug 2022 Bukhara, Uzbekistan
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- Developed optimal methods in collaborating with senior engineers to streamline sorting processes
- Contributed to redesigning the sorting station by applying optimization methods, resulting in a 35% cost reduction
- Performed maintenance for traffic lights and control relays in the sorting station
- Demonstrated strong problem-solving skills and an ability to adapt to challenging technical issues

Research Projects

Differential Games

- Working on slower Target and Defender against faster Attacker game
- Conducted rigorous quantitative analysis to find agents' optimal strategies
- Utilized MATLAB to simulate optimal strategies, visualizing dominance regions and trajectories
- Used control theory, optimization, and game theory to conduct this research

Reinforcement Learning for LQR Problem

- Investigated research papers on optimal control, on-policy, and off-policy Reinforcement Learning
- Analyzed and adapted DDPG, Twin Delayed DDPG, and Soft Actor-Critic algorithms for continuous control tasks
- Conducting in-depth research to improve the safety of autonomous vehicles and drones

Skills

Programming Languages: MATLAB, Python, C++

Software: Simulink, Arduino, CAD, Microsoft Office (Word, Excel, PowerPoint, Visio)

Packages/Frameworks: Numpy, Matplotlib, SciPy, Anaconda, Pandas, PyTorch, TensorFlow, GitHub

Soft: Problem-solving, Critical thinking, Leadership, Team collaboration, Communication, Creativity

Publications

“METHODS FOR INCREASING THE CAPACITY OF SORTING PROCESSES ON RAILWAY SORTING HILLS”,
Z. Toshboev, U. Abdullaev, Universum, Dec 27, 2022